Project Planning Document

Gait Recognition for User-Independent Access Control

Final Year Project 2024 COMP30151/52

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1 – Introduction

Authentication and authorisation are core security concepts that allow us to assert that an entity is who they claim to be and meets entry criteria (Sandhu and Samarati, 1994). With "more than nine in ten" organisations suffering social-engineering related compromises in 2023, defending against unauthorised access in any form is critical (Barracuda, 2024).

Many physical access control systems require users to scan a key card or other physical token to access an area. This only provides a single factor of authentication however; if no other checks are performed (Multi-Factor Authentication) – there is a risk a malicious actor gains unauthorised access using a stolen key card (IFSEC Global, 2024).

Another concern with physical access control authentication is usability. Between 2022 and 2024, the number of organisations adopting smartphone-based authentication rose from 32% to 39%. This indicates a move away from key card-based access control, to reduce issues caused by card loss/damage, and the threat of unauthorised access (IFSEC Global, 2022). This does not eliminate usability concerns however, users who wish to authenticate are still required to carry a physical token (a phone); there remains room for improvement which I believe can be achieved using biometrics.

This project aims to deliver a user-independent access control system that requires no interaction from users to authenticate. The primary deliverable will be a proof-of-concept door access control system that uses gait: the way a person walks, to identify the person approaching. This should help combat both concerns mentioned above: it's much harder to impersonate gait than it is to steal a key card, and a user can not only authenticate without carrying an authentication token; they don't have to interact with the system at all – a two-point convenience improvement (Whittle, 1996).

A sensor of some kind will be used to track movement and identify people approaching a doorway. This will require training of a machine-learning model using either previously collected movement data or a new dataset, created by conducting primary research. Once the recognition aspect is working, a modular design for the overall access control system can be introduced, where a central server is responsible for all more demanding data processing and identification in response to data collected by nodes.

Many existing gait analysis solutions capture multiple angles of a person's movements to assess how they walk; capturing someone's walking from the side for example may prove useful, allowing assessment varying attributes such as arms and legs moving back and forth (Singh *et al.*, 2018). The plan is to implement a single-view gait recognition algorithm which may affect identification confidence, but based on other research conducted in this area, can serve as a plausible and cost-effective proof of how a user-independent access control system can be implemented (Goffredo, Carter and Nixon, 2008).

The result of this project may prove useful in other areas too. For example, CCTV footage could be analysed and a criminal identified based on the way their gait (Muramatsu *et al.*, 2013).

Overall, I am excited to start work on research and development and believe the product will be interesting.

2 – Aims and Objectives

The primary goal of this project is to deliver a physical access control system that uses gait recognition to identify and authenticate a person without them needing to manually intervene. An authorisation element will then decide if the identified user is allowed to access the area protected by the system or not.

To meet this goal, machine learning will be used to assess movement data collected using a camera or sensor. The machine learning model must be trained on suitable historic data to ensure it can perform comparisons and differentiate individual's gait. Once the machine learning aspect of the project is complete, remaining actions will require more traditional software functionality to build out the design of the overall access control system.

I aim for the resulting access control system to be distributed and modular, consisting of a central server responsible for high power operations (gait assessment) and one or more nodes responsible for collecting movement data away from whichever entry way they are attached to, before sending it off for processing. This design should enable extensibility and allow for more configuration capability.

To achieve these aims, several objectives must be met, some of which are dependent on other actions/objectives being completed before them. Table 1: Project Objectives details and justifies these objectives with target dates for completion.

3 – Tasks and Deliverables

To achieve this project's aims and objectives, several smaller tasks must be completed, both research and development based. There will be several deliverables over the course of the project as things such as a movement dataset must be acquired or produced, before the machine learning model is produced, before the final deliverable, an access control system, is produced.

Table 2: Project Tasks lists (in order) all tasks that must be completed over the next six months to achieve the goals outlined in 2 – Aims and Objectives.

Through completion of these tasks, I aim to produce the following deliverables:

1. A machine learning model capable of assessing how a person walks, assessing gait to identify them.
2. A distributed access control system topology using several computer devices and sensors.
3. Access control software to utilise the machine learning model and physical system topology.
4. A report summarising my findings, with a critical evaluation and suggestions for future development and improvements.
5. A presentation and demonstration of the project, its findings, and functionality for university assessors or others.

4 – Gantt Chart

Using the items outlined in 2 – Aims and Objectives and 3 – Tasks and Deliverables a Gantt chart: Figure 3: Task Time Plan Gantt Chart has been created outlining each individual task that needs completing alongside the date work on the task should start, and when it concludes. Tasks relating to other university modules are also included in this Gantt chart as it is important to consider other commitments during the timeframe (APM, 2024).

Time blocks in this Gantt chart are divided into university weeks, starting at week 13 (w/c October 21st 2024) and ending at week 42 (w/c May 12th 2025). A month row also exists to try and reference the month these weeks fall inside of as best as possible.

This Gantt chart should serve well as a general time plan for the year. Using this to track tasks and deliverables and be able to allocate time will help prevent becoming overwhelmed by leaving things too late or forgetting to complete certain tasks. Depending on issues encountered, the contingency plan outlined in 6 – Risks may need following and adjustments made.

5 – Resources

To achieve this project's aims and objectives, a number of both software and hardware resources will be required, as well as research-based resources like research papers and historic datasets. The following list outlines resources that will be needed by requirement:

5.1 – Aim #1 – Machine Learning Resources

Several hardware and software resources will be needed when sourcing data, training, and testing a machine learning model. Table 3: Machine Learning for Gait Recognition Resources details and justifies what will be needed.

5.2 – Aim #2 - Access Control System Resources

After suitably training a machine learning model for gait recognition I must implement the access control system that uses the machine learning model to recognise gait and grant access. Table 4: Access Control Resources outlines the software and hardware resources that are likely required for this.

5.3 – Research and Information Based Resources

Aside from hardware and software required, I will require information such as datasets, historic research, and other documents to support my research and the completion of this project. Learning from these resources will be critical for my development. Table 5: Research and Information Resources lists what may be needed.

5.4 – Miscellaneous/General Resources

Throughout this project I will require several (mainly software) resources for general research and development. For example, an IDE for development, a computer, storage and more. Table 6: Miscellaneous Resources details what will likely be needed.

6 – Risks

There are several potential risks to the success of this project, some of which more likely to occur than others. Using Table 7: Risk Assessment Rating Matrix to score the probability a risk will occur and the impact it will have on the project from 1 – unlikely/insignificant to 5 – certain/critical, the risks related to this project have been listed and scored in Table 8: Risk Assessment Scoring (SafetyCulture, 2022).

There are several ways to mitigate risk, we can accept the risk may happen and continue without making any changes, make changes to avoid the risk completely, removing the potential it occurs, or produce plans to combat the risk in the event it happens. Table 9: Risk Contingency Planning lists all risks in Table 8 alongside a description and mitigation plan of what actions will be taken to avoid, accept, or combat the risk.

7 – Career Aspirations

I have several interests within computing, and while I have not yet settled on a specific field I would like to go into after my studies, I have key interests, some of which have been influenced by my year in industry.

My hope is that completing this project will allow me to advance my skills in areas such as machine learning, software development, cyber security, distributed systems, and high-performance computing. Below is an assessment of how this project could help influence my career opportunities in each of the fields:

7.1 – Computer and Information Security Fields

This project has an involvement in security as it aims to produce an access control system using a lesser used form of authentication, proposing an innovative solution to concerns about the most common forms of access control systems.

UK Government data indicates while demand for security professionals has dropped, the number of security graduates has risen by 34% (GOV UK, 2024).

The following security-related technical skills should be developed because of this project which may help me stand out: *Authentication and Authorisation knowledge, Access Control Methods, Python Programming, Anomaly Detection, Policy Design and Configuration.*

This project may appeal to employers looking for security professionals as it demonstrates knowledge and experience in access control. For example, a fire and security engineer job (https://careers.g4s.com/en/jobs/fire-and-security-engineer/74513) by a company called G4S asks for experience with security/alarm systems, somewhat relating to the context of this project (G4S, 2024).

7.2 – Machine Learning and High-Performance Computing Fields

The machine learning and distributed computing aspects of this project also excite me when it comes to progressing in my career.

During the second half of my placement year, I was in a team responsible for helping to develop machine learning algorithms for engineering R&D. Extending my knowledge in this area may enable me to better contribute if returning to the same organisation, or otherwise significantly contribute to my CV with skills such as: *Creating and Training*

Machine Learning Models, Creating Datasets, Python Programming, High Performance Computing, Complex Algorithms, Data Science/Analysis.

I had also been considering pursuing postgraduate education in the High Performance Computing or Data Science field with several programmes available at universities in the UK, including Edinburgh University (The University of Edinburgh, 2024). Whilst self-study during this project will greatly improve my skills, I understand the benefits of further study in a formal environment.

7.3 – Software Engineering and Development Fields

This project will consist of a significant amount of software development, mainly using the Python programming language, but also potentially JavaScript if necessary. I will be able to greatly expand my knowledge and experience in programming over the next six months when completing this project.

Having already spent time in software engineering during my year on placement, this project should both use and compliment my skills gained there.

I should be able to update my CV with details of a minimum of the following: *Python, JavaScript, Web Frameworks, Communication Protocols, APIs, Hardware Interfacing, Computer Vision, Raspberry PI development*.

A job posting by Mercedes AMG Petronas Formula 1 team for a “graduate race strategy engineer” requests experience with Python among other programming languages. Machine learning experience may also be valuable in this field based on knowledge of motorsport engineering careers (Mercedes-AMG PETRONAS F1 Team, 2024).

8 – Legal, Social, Ethical, and Professional Issues

8.1 – Legal Issues (Data Protection)

Laws such as the GDPR in the EU, or Data Protection Act in the UK exist to ensure best practices are followed when working with personal data (GDPR INFO, 2024).

As this project involves potentially collecting and working with personal biometric information (movement data) these laws must be adhered to and data stored securely, only used for its stated purpose (communicated in participant consent forms) and, when no longer needed, disposed of.

In the GDPR, biometric data is specially categorised if being processed to uniquely identify individuals. This purpose of this project is to do just that, therefore, explicit consent is needed for use of the data (or published with consent) (ICO, 2024).

Although cameras/sensors will be used to capture information, resulting datasets will not consist of images but coordinate data points, somewhat anonymising data and making it not personally identifiable which helps towards meeting these legal expectations.

8.2 – Social Issues

Social issues in the context of a machine learning project generally involve the privacy of subjects whose data will be used and the impact the resulting system may have on a community or in a workplace (Bacciu *et al.*, 2019).

8.2.1 – Surveillance and Privacy

Although the intended purpose of this system has good intentions, the technology produced could be repurposed and used for surveillance. For example, an organisation may be able to set this up to recognise individuals wherever they are, to automatically track how long someone spends in the cafeteria for example.

This could also be used to identify users in areas they aren't permitted, if a record is kept of who is allowed in what room depending on where sensors/cameras are setup.

The privacy concerns resulting from this can create a sense of uneasiness with users wondering what their data is being used for and when and where they are being tracked. This somewhat relates to the concept of the panopticon, a prison design where inmates are under constant observation by an entity they cannot see, resulting in less people acting out or breaking rules (Ball, Haggerty and Lyon, 2012).

8.2.2 – Accessibility and Usability

This system uses gait to uniquely identify individuals, but what happens if a disabled person, who cannot walk, needs to authenticate?

Situations like the above must be considered and the system must be designed in a way where fail safes exist (e.g., alternate methods of authentication) that still allow people unable to take advantage of core functionality to access what they need.

8.3 – Ethical Issues

Combatting ethical issues in the context of a machine learning project generally involves eliminating bias, acquiring consent to use human's data, and preventing misuse (Raji *et al.*, 2020).

8.3.1 – Machine Learning Bias

Some forms of machine learning-based biometrics have well known biases, for example, face recognition technology often struggles to differentiate individuals with darker complexions (Raji *et al.*, 2020).

In the context of this project, differences such as sex, age or disabilities could influence how someone walks and thus how their gait is assessed. If primary research is conducted, it is a good idea to collect a diverse set of data to train the machine learning model to hopefully combat this issue, making it more likely for recognition to work as intended.

8.3.2 – Risk of Misuse

Relating to 8.2.1 – Surveillance and Privacy, there is a risk that the technology produced because of this project would be misused. If implemented in a corporate environment, the most likely case of misuse would be the surveillance of employees. Depending on the level of transparency organisations have, this could be further complicated by data being collected without consent. As mentioned in 8.1 – Legal Issues (Data Protection) biometric data collected could be of special classification if it can be traced back to a person, if this is the case, organisations must acquire explicit consent for data collection.

In other contexts, this type of technology may be misused to monitor individuals in specific economic areas, perform mass surveillance, monitor gait-related medical issues, or make decisions based on gait data without human intervention (Kindt, 2013).

8.4 – Professional Issues

The BCS Code of Conduct outlines four key principles on producing work that best serves public interest, ensuring high work is produced to a high standard (BCS, 2024).

8.4.1 – Public Interest

The technology resulting from this project could have several impacts on society both positive and negative. For example, introducing gait recognition can help improve security overall – a positive. However, as mentioned in 8.2 – Social Issues, the potential to monitor activity covertly could make people uneasy. Discrimination (intentional or unintentional) could happen because of improperly calibrated biometric systems; targeting a certain demographic of individuals in a community with no accountability should not happen.

To act within public interest in mind, subjects whose data is collected by this system must be explicitly informed about what data is collected and given the option to opt out. The potential for inaccuracies or errors must be accepted and the system owner must accept responsibility for any issues (e.g., data protection, discrimination).

8.4.2 – Professional Competence and Integrity

It is critical that sufficient research is conducted throughout this project to ensure work produced is of a high standard. The gait recognition and access control system should be robust and reliable, testing must be conducted based on expectations, and the accuracy of the machine learning model should be regularly evaluated to check for errors leading to possible misidentification or discrimination.

In addition to ensuring system accuracy and reliability, the integrity and security of any data used or produced by the system must be kept. Reliable and secure data storage must be used to prevent data loss or unauthorised access.

Both the creators and implementers of the resulting access control system must have adequate knowledge of the data privacy laws associated with the types of information collected by the gait recognition system – discussed in 8.1 – Legal Issues (Data Protection), not having this knowledge could lead to accidental breaches in regulation, affecting any of the individuals whose data had been captured and potentially access maliciously.

8.4.3 – Duty to Relevant Authority

When developing his project, respect in reference to Nottingham Trent University is a must, breaches to data protection regulations or other ethical violations may affect the institution as well as the project owner (BCS, 2024).

Any concerns related to the legal handling of data, or to ethics must be raised to the most appropriate authority to ensure best practices remain followed and laws are not broken, this may be done through communication with the project supervisor who gives guidance throughout and may be able to identify unforeseen risk and advise on next steps.

In relation to the gait recognition system and collection of data through primary research and testing, subjects must be informed and give consent for their data to be collected, if there are further concerns with this, the relevant authority can advise/assist.

8.4.4 – Duty to the Profession

As a computing professional, the project owner must operate with integrity and be as transparent as possible. The intended uses of the gait recognition system must be clearly stated and not strayed from, and any data collected must only be used for the stated purpose. This project will consist of a significant amount of research, including the use of previously published papers. To contribute back, new findings should also be published for use by others, if possible, with credit and references used where applicable.

Responsibility and accountability for actions taken throughout the project belong to the project owner, thus any issues relating to breaches in data protection or improper ethics would be traced back. Operating with full transparency may curb any influence to change aspects of the project in bad faith; either due to a conflict of interest or for other reasons.

6 – Figures

Table 1: Project Objectives

#	Objective	Awaiting
1	Acquire hardware resources to enable collection and assessment of gait data (aim #1) by the end of October 2024. *This will enable the achievement of the first project aim: allowing gait information to be collected before being either used to train a machine learning model or passed through the machine learning model for evaluation and identification. These resources include:* - *A camera/sensor device (e.g., Webcam, Xbox Kinect)* - *A tripod/mount allowing the sensor to be easily positioned* - *Any I/O media needed for the sensor to connect to a computer*	
2	Acquire a dataset of movement data compatible with the hardware that is being used for this project by mid-November 2024. *This data set does not have to be huge for a proof-of-concept system and if collected using primary research should need a minimum of 10 participants to collect data from.* *Identifying users based on gait requires a model trained on similar, historic data that can perform comparisons and find the best (or no) match based on an individual's movement data.*	#1
3	Train a machine learning model on the acquired dataset using Python by mid-December 2024. *This is a core part of the system and is the first deliverable I aim to produce, enabling identification of individuals based on an evaluation of their gait. My hope is the dataset used to train the model will be sufficient to prove the capability of gait-based identification.*	#2
4	Test the trained machine learning model to ensure it is capable of identifying and matching captured gait data to data from the dataset by the end of December 2024. *This is critical as a model that doesn't work will make it impossible to identify individuals based on their gait, meaning I can't achieve what I set out to do in producing an access control system that operates without user interaction.*	#3
5	Acquire hardware resources to enable a move to a distributed system design where movement data is sent from a remote node to a server for evaluation my mid-January 2025. *After a suitable model has been trained and can identify individuals based on gait. The second part of the system can be created: the distributed access control system. This will require a minimum of the following hardware for demonstration purposes:* - *A main computer (central "server" device)* - *A secondary computer (remote "node" device)*	#4

	- *Networking equipment to connect the computers* - *Small scale output, actuator, or door control*	
6	Implement the distributed system design with all hardware and software and functionality to produce a working gait-based access control system with relevant configuration by the end of February 2025. *This uses the previous deliverable of a trained machine learning model to provide a gait-based access control system. Distributed design should make the system more extensible and better justify an as-is implementation in multiple places across a workplace, for example.* *It is important for some form of output to be given to demonstrate access being given. For the proof-of-concept this doesn't have to be a physical door but could be a smaller scale simulation.*	#3, #5
8	Complete testing and evaluation of the system and the project report by the start of April 2025.	#6

Table 2: Project Tasks

#	Task	Awaiting
1	Identify, evaluate, and acquire suitable sensor equipment to capture gait	
2	Test sensor equipment to ensure measurable data can be collected	1
3	Create relevant participant information and informed consent documents for use in primary research.	2
4	Hold sessions to collect movement data of individuals using the sensor to create a dataset consisting of several people's movement data	3
5	Evaluate machine learning models to identify the most suitable for the use case of comparing gait data and identifying individuals (e.g., SVM)	4
6	Train a machine learning model on the acquired gait movement dataset using Python	5
7	Test the trained machine learning model to confirm it is suitable for identifying individual's gait patterns, evaluate and document the performance of the model	6
8	Adjust and make changes to the machine learning model based on results of testing, if needed	7
9	Identify, evaluate, and acquire suitable computer and networking equipment to implement the distributed access control system design	
10	Identify the most suitable software technologies to string together the hardware and machine learning model in a network to achieve the distributed design	9

11	Physical topology of the distributed access control system design that uses the previously trained machine learning model to evaluate gait and grant access based on authorisation	
12	Gait data capture algorithm on the node device to capture movement data points using the hardware sensor	10
13	Machine learning model on the server device, with an API feature to receive data from nodes and respond based on whether the identified user is authorised	10
14	Add output on the node device to simulate access being granted in response to an OK message from the server if a person is authorised	12, 13
15	Test the overall system to confirm it works as intended, meaning that it must be able to identify a user by gait (if they are in the datastore) and give output to confirm the door would unlock if they were authorised	14
16	Evaluate the solution against initial aims and objectives and propose future development ideas and improvements	15
17	Summarise findings, development and testing process, and improvements in a report document alongside literature review of issue	

Table 3: Machine Learning for Gait Recognition Resources

#	Resource	Description/Justification
1	Video Camera or Movement Sensor	To capture subject's movement, an adequate sensor is required that will capture data in a high enough resolution to perform operations, train a machine learning algorithm, and evaluate using said algorithm to uniquely identify a subject. Based on the approach I take to gait recognition; I have two main choices for this: Xbox Kinect V2 Sensor (Figure 1: Xbox Kinect V2 Sensor**Error! Reference source not found.**) The Xbox Kinect V2 sensor is an incredibly powerful piece of consumer hardware, traditionally used for gaming. However, with an in-built depth sensor, 2GBps data processing capability, and "joint mapping" I think this could prove effective at providing exactly the right data to train a gait recognition machine learning model (Michael *et al.*, 2017). The Kinect V2 sensor (https://uk.webuy.com/product-detail?id=SXB1OFFKIN2) and adapter needed to connect to a computer (https://uk.webuy.com/product-detail?id=0885370862478) come out to around £40 second hand, a cost I think is justifiable given their usefulness. Logitech 1080p Webcam (Figure 2: Logitech C920 1080p Webcam) A more traditional gait recognition method may be possible using just a camera, operations could be performed on a video feed of a webcam to separate the subject from the background and assess their movement. Historic research data and videos of humans walking may be better available for this (Goffredo, Carter and Nixon, 2008).

2	Tripod or Sensor Mount	To ensure the sensor can be positioned correctly, a tripod or mount would prove useful. I already have one of these, capable of extending to a maximum of six feet which will be enough (https://www.manfrotto.com/uk-en/element-mii-aluminium-red-mkelmii4rd-bh/).
3	High Powered Computer	A computer powerful enough to support machine learning and other developments for this project is required. I have a personal computer I may use and may also be able to use university computers to support me with this.
4	Programming Language	To create/train a machine learning model, a suitable programming language will be needed. I plan to use Python (https://www.python.org/) throughout this project to achieve this.
5	Machine Learning Library	There are several machine learning libraries compatible with Python that provide several model types and functions to allow the creation of a machine learning model that should help me achieve my project aims. (e.g., Scikit-learn, TensorFlow, and PyTorch)

Table 4: Access Control Resources

#	Resource	Description/Justification
1	Node Computer	To enable the distributed system design, one or more node device will be required and will be responsible for collecting movement data and sending it to the server/high powered computer. I plan to use a Raspberry PI device (https://www.raspberrypi.com/products/raspberry-pi-3-model-b/) to do this which can be provided by the university.
2	Networking Equipment	Creating a distributed topology for proof-of-concepts sake should be rather simple. Hardware required for this would be a small, unmanaged switch and some Ethernet cables at most, if not only an/some Ethernet cables.
3	Small-Scale Simulation Output	To provide output and simulate a door opening when access is given by the access control system a small-scale output can be used. For example, connecting an LED light to the node (Raspberry PI) that lights up when a qualifying gait is detected is one option.
4	Programming Language(s)	I aim to keep a simple software stack for this project, therefore continuing with Python (https://www.python.org/) is my best option, this language should provide enough functionality to implement networking and communication across devices to transfer gait data and respond with messages instructing nodes to give access or not.

5	Communication Protocols and Libraries	To communicate across devices using Python, several protocols can be used. For example, using the Socket library (https://docs.python.org/3/library/socket.html) would allow for low-level UDP or TCP communication. However, using HTTP with a library like Django (https://www.djangoproject.com/ may prove more robust.

Table 5: Research and Information Resources

#	Resource	Description/Justification
1	Historic Datasets for Research and Training	Historic datasets may prove useful to learn how movement data is structured, to test training a machine learning model on certain data, or even to reduce workload if a suitable enough dataset can be found that removes the need to conduct primary research. Platforms such as Zenodo and Figshare can be used for this as other researcher may have shared relevant information that proves useful to this project.
2	Implementation research	Research completed in the field of gait analysis and machine learning will prove useful for this project, as learning from other researcher's findings may both help me to progress and improve upon their findings if holes in literature are found to produce a better result. Platforms such as Google scholar and IEEE xplore can be used to find relevant papers for this.

Table 6: Miscellaneous Resources

#	Resource	Description/Justification
1	Software IDE	To develop the software for this project I will need to use a suitable IDE. I plan to use Visual Studio Code with this as it is compatible with Python and should provide all the functionality I need.
2	Computer and Peripherals	A computer and suitable peripherals will be required throughout this project both to write the report and work on software development. I plan to use my personal and university devices to do this.
3	Digital Storage	I will require adequate storage to store the machine learning models, data sets, code, and documentation. I plan to use my computer or OneDrive for this.

Figure 1: Xbox Kinect V2 Sensor

Figure 2: Logitech C920 1080p Webcam

Figure 3: Task Time Plan Gantt Chart

	Task	Start (Week)	End (Week)	Duration (Weeks)
1	**Aim #1 - Gait Recognition Machine Learning Model**	**13**	**22**	**8**
	Acquire suitable gait capture sensor equipment	13	14	2
	Test acquired sensor equipment	14	14	1
	Create Participant Information and Informed Consent documents	15	16	2
	Primary research data capture sessions to create gait datasets	15	16	2
	Identitfy best machine learning model type to use for project	17	20	4
	Train machine learning model on acquired gait dataset(s)	17	20	4
	Test and correct the trained machine learning model	21	22	2
	Evaluate and document the performance of the machine learning model	22	22	1
2	**Aim #2 - Access Control System Using Machine Learning Model**	**23**	**31**	**9**
	Complete a logical design of the distributed topology	23	23	1
	Acquire suitable computing and network hardware to implement the topology	23	24	2
	Identify the most suitable software stack to use to implement the access control system	25	25	1
	Implement the physical rendition of the distributed topology	25	26	2
	Create the gait data capture algorithm on the node device	27	30	4
	Create server side API (inc ML model) for gait analysis requests from node devices	27	30	4
	Connect output (door, light, etc) to node computer to indicate successful authentication	30	30	1
	Complete testing of overall access control system with multiple subjects	31	31	1
3	**Overall Project and Reporting**	**32**	**38**	**7**
	Complete a critical evaluation of the final design and document findings	32	36	5
	Summarise project with a literature review, implementation steps, and evaluation in report	34	36	3
	Present and demonstrate project findings at university event	38	38	1
4	**Other Modules / Comittments**	**13**	**42**	**32**
	Security in Practice Coursework 1	13	17	6
	Digital Investigations Assessment 1	14	20	7
	Security in Practice Coursework 2	25	35	11
	Advanced Topics in Cyber Security Coursework + Presentation	13	36	25
	Digital Investigations Assessment 2	32	40	9
	Internet of Things Report	29	39	11
	Internet of Things Exam	41	42	2

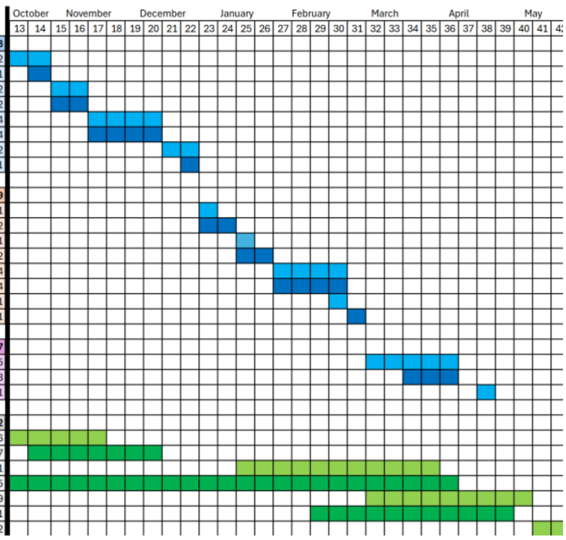

Table 7: Risk Assessment Rating Matrix

		Impact				
		Insignificant 1	Minor 2	Significant 3	Major 4	Severe 5
Probability	Almost Certain 5	Medium 5	High 10	Very High 15	Critical 20	Critical 25
	Likely 4	Medium 4	Medium 8	High 12	Very High 16	Critical 20
	Moderate 3	Low 3	Medium 6	Medium 9	High 12	Very High 15
	Unlikely 2	Very Low 2	Low 4	Medium 6	Medium 8	High 10
	Rare 1	Very Low 1	Very Low 2	Low 3	Medium 4	Medium 5

Table 8: Risk Assessment Scoring

Area	#	Risk	Probability	Impact	Rating
Technical Feasibility	1	Machine Learning Knowledge and Familiarity	3	5	Very High 15
	2	Distributed Computing and Networking Familiarity	2	3	Medium 6
	3	Project Size and Scope being too large/wide	2	3	Medium 6
	4	Hardware Functionality and Compatibility	3	5	Very High 15
	5	Computer Processing Power Issues	1	4	Medium 4
	6	Integration Challenges with Access Control System	2	4	Medium 8
Information Accessibility	7	Insufficient data capture or dataset resolution	2	4	Medium 8
	8	Limited historic research or dataset availability	2	3	Medium 6
	9	Unsuitable machine learning models for this use case	2	5	High 10
	10	Unsuitable speed of algorithms for smooth operation	2	2	Low 4
	11	Data Processing Liability (Legality Related)	1	5	Medium 5
Procedural Risks	12	Issues with subject availability for data collection	2	4	Medium 8
	13	Issues with performing adequate tests to assess functionality	3	3	Medium 9
	14	Unforseen delays to project for personal reasons	2	5	High 10
	15	Unforseen availability issues with project tutor	2	2	Low 4
	16	Unforseen data loss of documentation, report or code	1	5	Medium 5
	17	Exceeding Timeline due to workload	2	4	Medium 8

Table 9: Risk Contingency Planning

#	Risk Description	Acceptance / Mitigation / Avoidance Plan
1	My knowledge of machine learning as it stands is somewhat limited. This could cause issues during the project as I may possibly struggle to identify a suitable machine learning model or fix an issue during training the model.	The solution to a lack of knowledge is research. If an issue is encountered, I will likely be able to find a solution online or in historic research papers. I am also able to contact university staff who may have more experience for guidance.
2	Familiarity with distributed system and network design and architecture is critical to deliver the access control system aim discussed in 2 – Aims and Objectives.	My familiarity with distributed systems, networks, and communication protocols is good. I don't expect this to be an issue. However, if issues are encountered, additional research can be conducted, and the project supervisor or other university staff can be consulted and asked for advice. If I am unable to overcome issues related to this risk, the project scope may change to allow development to continue (e.g., making the access control system operate on a single device)
3	The scope/size of the project may exceed my capabilities or what is necessary to create an access control system with the expected functionality.	If experiencing scope creep, or issues with not being able to complete all expected objectives within the same project, the scope can be narrowed or method of implementation changed (e.g., using a different form of biometrics, only delivering the machine learning model instead of implementing it into a distributed system, etc...)
4	Hardware used for this project: the computer, raspberry PI, camera/sensor, and network equipment may have compatibility issues (although not likely)	If compatibility issues are encountered with the acquired hardware, which in the most likely case will be compatibility of the sensor and computer(s), I will need to identify an alternate sensor that can be used and still provide enough data, at a high enough resolution to meet the project goals. If a Kinect sensor is used, the issue may be with the adapter used to connect it to the computer, in this case a different adapter could be tested before trying a different sensor type.
5	Computers used to train the machine learning model, or to use the model to identify gait may not have sufficient processing power.	If the personal computer(s) I am using for development prove to have insufficient power, I will enquire about using the high-power computers in the university HPC lab. These computers I know are more than capable of training machine learning models. The

		compromise is that I complete all development on-site.
6	After the aim of creating a machine learning model for gait recognition is complete, issues may be encountered when integrating it into an access control system. For example, transferring data between devices over a network, receiving responses in adequate time, etc.	As I aim to deliver a proof-of-concept, speed of processing is not the most critical aspect. If issues with integration of the two parts of this project are encountered that don't relate to transfer speed of data, adjustments could be made to either simulate the operation of the access control as well as proving the machine learning model worked independently, or the scope of the project could be changed to deliver only the machine learning model.
7	The acquired dataset may be insufficient and contain several subjects too low to train the model effectively.	Additional primary research can be conducted to increase the size of the data set first-hand. Alternatively, historic research data can be looked up on sites such as Figshare if available and I can attempt to use it for this research. If these datasets remain insufficient, the method of biometrics may need to change to something like face recognition where datasets are much simpler and machine learning models easier to train.
8	A limited amount of historic research papers and datasets may cause delays in development as nothing can be referenced to follow best practice or use known, good quality data for model training.	If this is the case, I may have to conduct primary research and collect/create my own datasets as soon as possible. This will allow me to experiment with the data and find the best way to capture and compare walking data to both train and use the machine learning model. Identifying if this may be an issue early is important as I'd have to start primary research as early as possible to trial an error and be in the best standing.
9	Existing machine learning models may be unsuitable for this use case. It may not be possible to train a machine learning model on movement data to assess gait (although unlikely)	If this is the case, it would be best to pivot to a different form of biometric with data that can be used to train a machine learning model, such as face recognition.
10	The speed of the machine learning model/algorithms may impact how quickly a user is authenticated and must wait to be "granted access"	As this is a proof-of-concept this is not too much of a concern so I think this risk can be accepted, efficiency improvements can be researched to help improve the speed if time allows, otherwise this may be included in things to improve mentioned in the report produced at the end of the project.

11	Issues surrounding the legality of collecting and processing human movement data may arise.	If primary research must be conducted, I plan on having all participants fill out consent forms after communicating what data will be collected and what it will be used for. I plan to keep this data confidential and will only use it for the intended purpose of training the machine learning model. We are avoiding this risk by following the correct process.
12	There may be a limited availability of subjects to conduct primary research with. The number of people available to collect movement data from.	This is unlikely due to the number of attendees at the university who may be available to contribute. If there is a lesser number of participants, a smaller dataset may have to be used, I may have to reach out to more people to try and collect data, or I may have to rely more on finding existing historic datasets online. If all else fails and I am unable to collect any datasets, I may have to pivot to a different form of bio-authentication.
13	Issues encountered during testing may prove the model does not work as intended.	If this is the case, further research will need to be conducted to see how the model can be improved to correctly recognise gait. Further development and subsequent testing will be needed to prove this out. If testing continues to fail, increasing the size of the dataset used to train the model may help solve the issue. If this fails, a different form of biometric authentication may need to be used.
14	Personal circumstances may lead to unforeseen delays in the project if I am unable to complete work.	If this is the case, I should submit an NEC to the university requesting my deliverables be put on hold. If possible, I can continue working on the project, however if not, work can resume once personal issues are sorted.
15	My project tutor may become unavailable for a period of time during the project	As my project tutor is not required for work to be completed, the project may not come to a stop, however, as advice could not be received I may have to seek advice from another university tutor if possible.
16	During the project, data loss could happen anywhere, whether it be the dataset, project code, or report.	To combat this, I plan to backup all project data to cloud storage. This will ensure that in the event data on a computer is corrupted, It can be recovered from the internet. I also plan to use version control for the project code, meaning I can track and control changes to ensure smoother and less risk prone development.

17	Also relating to the scope of the project, If the workload is too much and the project size is too big, the timeline could be exceeded, and due date missed.	I plan to assess how this project progresses over the next couple of months. Based on whether the first tasks and objectives are completed in the expected timeframe, adjustments can be made. For example, If I begin to miss objective dates, I will have to assess the size of the project and potentially scale it down to a lesser deliverable to ensure that something tangible can still be produced by the deadline.

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